Wenjie Ji

Westlake University
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RESEARCH INTEREST

Quantum Matter, Highly Entangled Quantum Matter, Quantum Information

EMPLOYMENT

Westlake University

Associate Professor April 2026 - Present.

McMaster University

Assistant Professor October 2024 - March 2026.

Perimeter Institute for Theoretical Physics

Associate Faculty October 2024 - March 2026.

California Institute of Technology

Ultra-Quantum Matter Postdoc Fellow August 2023 - September 2024.

Kavli Institute for Theoretical Physics, University of California, Santa Barbara

Postdoc Scholar September 2020 - July 2023.

EDUCATION

Massachusetts Institute of Technology

Ph.D. Physics, 2020.

Thesis title: Anomalies and symmetries on the boundary of topologically ordered phases.

Thesis advisor: Xiao-Gang Wen

Tsinghua University, Beijing

B.Sc. Physics, 2012.

AWARDS AND HONORS

Natural Sciences and Engineering Research Council of Canada, Discovery Grant Award, 2025-2030

Natural Sciences and Engineering Research Council of Canada, Discovery Launch Supplement Award, 2025-2026

MIT Marble Presidential Fellowship, 2012-2013

Tsinghua University, Ye Qisun Award, 2012

Tsinghua University, National Endeavor Scholarship, Academic 1st level, 2011

"Friends of Tsinghua" Scholarship, Academic 1st level, 2010

Tsinghua University, National Endeavor Scholarship, Academic 1st level, 2009

Tsinghua University, Member of Tsinghua Xuetang Talent Program, 2009

PUBLICATIONS

[1] Wenjie Ji, Ryan A Lanzetta, Zheng Zhou, and Chong Wang. Self-dual higgs transitions: Toric code and beyond. *arXiv preprint arXiv:2601.20945*, 2026

[2] Wilbur Shirley, Carolyn Zhang, Wenjie Ji, and Michael Levin. Anomaly-free symmetries with obstructions to gauging and onsiteability. *Phys. Rev. Lett.*, 136:216602, May 2026

[3] Wenjie Ji, David T Stephen, Michael Levin, and Xie Chen. Bulk excitations of invertible phases. *arXiv:2506.11288*, 2025

[4] Wenjie Ji and Xie Chen. Topological defects of 2+1D systems from line excitations in 3+1D bulk. *arXiv:2407.02488*, 2024

[5] Arkya Chatterjee, Wenjie Ji, and Xiao-Gang Wen. Emergent generalized symmetry and maximal symmetry topological order. *Phys. Rev. B*, 112:115142, Sep 2025

[6] Wenjie Ji, Nathanan Tantivasadakarn, and Cenke Xu. Boundary states of three dimensional topological order and the deconfined quantum critical point. *SciPost Physics*, 15(6):231, 2023

[7] Shang Liu and Wenjie Ji. Towards non-invertible anomalies from generalized ising models. *SciPost Physics*, 15(4):150, 2023

[8] Jong Yeon Lee, Wenjie Ji, Zhen Bi, and Matthew P. A. Fisher. Decoding Measurement-Prepared Quantum Phases and Transitions: from Ising model to gauge theory, and beyond. *arXiv:2208.11699*, 2022

[9] Victor V Albert, David Aasen, Wenqing Xu, Wenjie Ji, Jason Alicea, and John Preskill. Spin chains, defects, and quantum wires for the quantum-double edge. *arXiv:2111.12096*, 2021

[10] Nayan E Myerson-Jain, Shang Liu, Wenjie Ji, Cenke Xu, and Sagar Vijay. Pascal's triangle fractal symmetries. *Physical Review Letters*, 128(11):115301, 2022

[11] Nathanan Tantivasadakarn, Wenjie Ji, and Sagar Vijay. Non-abelian hybrid fracton orders. *Physical Review B*, 104(11):115117, 2021

[12] Wenjie Ji and Xiao-Gang Wen. A unified view on symmetry, anomalous symmetry and non-invertible gravitational anomaly. *arXiv:2106.02069*, 2021

[13] Po-Shen Hsin, Wenjie Ji, and Chao-Ming Jian. Exotic invertible phases with higher-group symmetries. *SciPost Physics*, 12(2):052, 2022

[14] Nathanan Tantivasadakarn, Wenjie Ji, and Sagar Vijay. Hybrid fracton phases: Parent orders for liquid and nonliquid quantum phases. *Physical Review B*, 103(24):245136, 2021

[15] Xiao-Chuan Wu, Wenjie Ji, and Cenke Xu. Categorical symmetries at criticality. *Journal of Statistical Mechanics: Theory and Experiment*, 2021(7):073101, 2021

[16] Wenjie Ji and Xiao-Gang Wen. Categorical symmetry and noninvertible anomaly in symmetry-breaking and topological phase transitions. *Physical Review Research*, 2(3):033417, 2020

[17] Wenjie Ji and Xiao-Gang Wen. Metallic states beyond the tomonaga-luttinger liquid in one dimension. *Physical Review B*, 102(19):195107, 2020

[18] Wenjie Ji, Shu-Heng Shao, and Xiao-Gang Wen. Topological transition on the conformal manifold. *Physical Review Research*, 2(3):033317, 2020

[19] Wenjie Ji and Xiao-Gang Wen. Noninvertible anomalies and mapping-class-group transformation of anomalous partition functions. *Phys. Rev. Research*, 1:033054, Oct 2019

[20] Wenjie Ji and Xiao-Gang Wen. Continuous topological phase transition between two 1d antiferromagnetic spin-1 boson superfluids with the same symmetry. *Physical review letters*, 123(3):035301, 2019

[21] Wenjie Ji and Xiao-Gang Wen. Realizing majorana zero mode with a stripe of ionized and isolated magnetic atoms on a layered superconductor. *arXiv:1804.10198*, 2018

[22] Wenjie Ji and Xiao-Gang Wen. 1/2 (e2/h) conductance plateau without 1d chiral majorana fermions. *Physical review letters*, 120(10):107002, 2018

[23] Mark JH Ku, Wenjie Ji, Biswaroop Mukherjee, Elmer Guardado-Sanchez, Lawrence W Cheuk, Tarik Yefsah, and Martin W Zwierlein. Motion of a solitonic vortex in the bec-bcs crossover. *Physical review letters*, 113(6):065301, 2014

[24] Tarik Yefsah, Ariel T Sommer, Mark JH Ku, Lawrence W Cheuk, Wenjie Ji, Waseem S Bakr, and Martin W Zwierlein. Heavy solitons in a fermionic superfluid. *Nature*, 499(7459):426, 2013

SELECTED INVITED TALKS

University of Toronto, Toronto Quantum Matter Seminars, Toronto, Canada, October 2025. *Talk title:* Anomaly-Free Symmetries that cannot be realized On-site or Gauged.

Institute for Advanced Study, Tsinghua University, Workshop, "Frontiers of Quantum Matter", Beijing, China, August 2025. *Talk title:* Anomaly-Free Symmetries that cannot be realized On-site or Gauged.

Peking University, Conference, "Generalized Symmetries in HEP and CMP", Beijing, China, July 2025. *Talk title:* Low Entanglement Excitations in Invertible Phases.

Canadian Association of Physicists, Congress, "2025 CAP Congress", Saskatoon, SK, Canada, June 2025. *Talk title:* Non-anomalous, Non-onsitable Symmetry.

International Centre for Mathematical Sciences, Workshop, "Categorical symmetries in Quantum Field Theory", Edinburgh, Jun 2024. *Talk title:* Topological defects from bulk to boundary.

Kavli Institute of Theoretical Physics, Conference, "Theories, Experiments and Numerics on Gapless Quantum Many-body Systems", Santa Barbara, May 2024, *Talk title:* Low Entanglement Excitations in Invertible Phases.

Perimeter Institute, Conference, "Higher Categorical Tools for Quantum Phases of Matter", Waterloo, Mar 2024. *Talk title:* Bulk Excitations of Invertible Phases.

The Chinese University of Hong Kong, Colloquium, Hong Kong, Dec 2023. *Talk titile:* Symmetry Diagonosed from a Topological Bulk.

University of Sydney, Workshop, "Topological Order on the Pacific Ocean", Sydney, Nov 2023. *Talk title:* Boundary states of the three-dimensional toric code.

California Institute of Technology, IQIM Seminar, Pasadena, Oct 2023. *Talk title:* Measurement-prepared Quantum Phases and Transitions.

Nordic Institute for Theoretical Physics, Nordita Program, "Categorical aspects of symmetries", Stockholm, Aug 2023. *Talk title:* A construction of stabilizer Hamiltonians realizing topological / fracton orders and symmetries on their boundaries.

Aspen Center of Physics, Summer Workshop, "Physics and Information Capabilities of Highly Entangled Quantum Matter", Aspen, Jul 2022. *Talk title:* A construction of stabilizer Hamiltonians realizing topological / fracton orders and symmetries on their boundaries.

Perimeter Institute, Condensed Matter Seminar, Waterloo, Mar 2021. *Talk title:* Hybrid fracton phases: Parent order for liquid and non-liquid quantum phases.

California Institute of Technology, Condensed Matter Seminar, Pasadena, Feb 2021. *Talk title:* Generalized symmetries at criticality.

Perimeter Institute, Condensed Matter Seminar, Waterloo, Oct 2019. *Talk title:* Non-invertible anomalies and topological orders.

Station Q, Microsoft, Seminar, Santa Barbara, Oct 2019. *Talk title:* Non-invertible anomalies and topological orders.

Francqui Symposium, “Wavefunctions and Entanglement in Quantum Field Theory”, University Foundation, Brussels, Sep 2019. *Talk title:* Non-invertible anomaly and Topological order.

Center of Mathematical sciences and applications, Harvard University, Workshop, “Topology and Dynamics in Quantum Matter”, Cambridge, Sep 2019. *Talk title:* Non-invertible anomaly and Topological order.

ORGANIZED WORKSHOPS & CONFERENCES

Perimeter Institute, Conference, "Physics of Quantum Information II", Waterloo, Canada, May 25-29, 2026.

Simons Center for Geometry and Physics, Workshop, "Applications of Generalized Symmetries and Topological Defects to Quantum Matter", Stony Brook, US, September 9-13, 2024.

TEACHING EXPERIENCE

2025 Fall	Instructor. 3MM3 Quantum Mechanics I. McMaster.
2020 Spring	Teaching Assistant. 8.371 Quantum Information II. MIT.
2019 Spring	Teaching Assistant. 8.06 Quantum Physics III. MIT.
2018 Fall	Teaching Assistant. 8.06 Quantum Physics III. MIT.
2018 Spring	Teaching Assistant. 8.511 Theory of Solids. MIT.
2017 Fall	Teaching Assistant. 8.333 Statistical Mechanics I. MIT.
2016 Fall	Teaching Assistant. 8.06 Quantum Physics III. MIT.

SERVICE

Regular referee for Physical Review Letters, Physical Review B, Physical Review D.